

Admit: 03.01.2024

Discharge: 18.01.2024

Service: Hematology/Oncology

Attending: [Name], MD

Diagnosis: Acute Myeloid Leukemia, ELN 2022 Intermediate, NPM1 wild type with FLT3-ITD

First-line therapy: 7+3 + midostaurin — induction started 09.05.2023; now on post-remission therapy with HiDAC + midostaurin

Brief History

Ms. Bennett, a 61-year-old teacher, presented **02.05.2023** with fatigue and gingival bleeding. Admission **WBC $19.4 \times 10^9/L$** (blasts 33%), **Hgb 7.8 g/dL**, **Plt $49 \times 10^9/L$** , **LDH 540 U/L**. Bone marrow showed **64% blasts**; flow **CD34⁺/CD117⁺/HLA-DR⁺**; karyotype otherwise normal. NGS: **FLT3-ITD** positive (allelic ratio low–intermediate), **NPM1 wild type**, no IDH mutations.

Induction (09–15.05.2023): Standard 7+3 with **midostaurin 50 mg PO bid** on days 8–21. Course notable for one **febrile neutropenia** episode (cultures negative), grade 2 mucositis, and cytarabine conjunctivitis prevented with steroid eye drops.

Day-28 marrow (06.06.2023): Complete remission (CR), blasts <2%, **FLT3-ITD PCR low-positive** near assay limit.

Consolidation: HiDAC (3 g/m² q12h D1,3,5) + **midostaurin D8–21** planned for 3–4 cycles. She completed cycles in **July, September and November 2023** without neurotoxicity. **Peripheral blood FLT3 PCR** became **undetectable (10.2023)** and has remained negative.

She was admitted this January for **Consolidation Cycle #4** at our center after a brief family move; we aligned dosing with prior records.

Hospital Course

Regimen administered:

- **Cytarabine 3 g/m² IV q12h** on **Days 1, 3, 5**.
- **Midostaurin 50 mg PO bid Days 8–21** (dispensed for home continuation).
- **Prophylaxis:** **Acyclovir 400 mg bid**; antibacterial (**levofloxacin 500 mg qd**) to begin if ANC <0.5; antiemetics **ondansetron** and **olanzapine 2.5 mg qHS**; **dexamethasone eye drops** for 48–72 h after each cytarabine dose; **PJP prophylaxis** not indicated given anticipated duration of neutropenia and prior mold prophylaxis history. We **avoided azoles** during midostaurin days due to CYP3A interaction/QTc.

Monitoring: Daily neuro checks (finger-to-nose, tandem gait, rapid alternating movements) — **no cerebellar findings**. ECG baseline and Day 10 with **QTcF 447→452 ms**. Strict I/O; TLS markers stable.

Complications/Interventions:

- **Nausea** grade 1–2 controlled with scheduled ondansetron; olanzapine at night improved sleep and appetite.
- **Cytopenias:** Gradual decline; no transfusions needed inpatient (discharge **Hgb 9.6**, **Plt 82**×10⁹/L; see trend below).
- **Electrolytes:** Mild hypomagnesemia corrected to maintain **Mg²⁺ ≥2.0**; **K⁺** maintained ≥4.
- **Infections:** Afebrile throughout admission; surveillance cultures negative.
- **Glycemic control:** Steroid eye drops transiently raised fasting glucose (peak 146 mg/dL) managed with dietary measures.

Key Labs:

- **CBC Trend**
 - 03.01: WBC 3.8 (ANC 1.4), Hgb 10.4, Plt 128
 - 14.01 (nadir): WBC 2.2 (ANC 0.7), Hgb 9.8, Plt 96
 - **18.01 (DC): WBC 2.6 (ANC 0.8), Hgb 9.6 g/dL, Plt 82**×10⁹/L
- **CMP (18.01):** Na 140, K 4.1, **Cr 0.90 mg/dL**, AST/ALT **24/21**, Tbili 0.6, albumin 3.9.
- **LDH:** 176 U/L.
- **FLT3-ITD PB PCR (01.2024): Undetectable.**
- **Echocardiogram (01.2024): LVEF 60%.**

Imaging: Not indicated; prior CXR (01.2024) normal.

Pathology/MRD history:

- Marrow (10.2023) MRD negative by **FLT3-ITD PCR** and **flow**; **repeat PB PCR 12.2023** remained negative. No new mutations on surveillance NGS (PB).

Discharge Condition

Stable, afebrile, ECOG **0–1**. Ambulates independently. No mucositis or ocular complaints. PICC site clean/dry/intact.

Discharge Medications

- **Midostaurin 50 mg PO bid** through **Day 21** of the cycle (counseled on food requirement and adherence).
- **Acyclovir 400 mg bid** (continue).
- **Levofloxacin 500 mg qd only if** ANC <0.5 per clinic instructions; patient will be called when to start based on labs.
- **Ondansetron 8 mg q8h PRN**; **olanzapine 2.5 mg qHS** x 7–10 days if nausea persists.

- **Dexamethasone ophthalmic** drops as prescribed during cytarabine window (completed inpatient).
 - **Home meds:** amlodipine 5 mg qd, levothyroxine 50 µg qAM, pravastatin 20 mg qHS, famotidine 20 mg PRN, acetaminophen PRN.
 - **Avoid** grapefruit/Seville orange and **all azole antifungals** while on midostaurin (CYP3A interaction/QTc).
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Comorbidities & Allergies

- **Hypertension, primary hypothyroidism, hyperlipidemia** (pravastatin chosen for minimal interactions), **GERD**.
 - **NKDA**.
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Follow-up & Ongoing Plan

1. **Labs:** CBC/CMP at the day-hospital **twice weekly** for 2 weeks, then weekly as counts recover. Transfuse per thresholds (Hgb ≥ 8 , Plt ≥ 10 ; ≥ 20 if bleeding). Consider **pegfilgrastim** at Day 22 if ANC < 0.3 or slow recovery.
2. **Infection precautions:** Call for T ≥ 38.0 °C, rigors, dyspnea, new cough, dysuria, or any bleeding.
3. **ECG/QTc:** Repeat in **3–4 weeks** or earlier if palpitations/syncope; maintain **Mg²⁺ ≥ 2.0** and **K⁺ ≥ 4.0** .
4. **MRD surveillance:** **Peripheral blood FLT3-ITD PCR** with each consolidation cycle and at end-of-therapy; **bone marrow** only if cytopenias persist, MRD rises, or counts fail to recover.
5. **Next step:** Treatment completed – transition to surveillance, as transplant is not routinely indicated for her risk category in sustained MRD-negative remission. Discuss oral Azacitidine maintenance.
6. **Supportive care:** Continue eye lubrication for 48 h after any future cytarabine doses; avoid contact lenses during that window; neuro checks to continue at home (report dysmetria, ataxia, dysarthria).
7. **Lifestyle:** Walking program 30 min/day; vaccinations per schedule (influenza annually; COVID updated booster).

When to call immediately: Fever ≥ 38.0 °C, chest pain, resting dyspnea, confusion, uncontrolled vomiting, **bruising/bleeding**, or sudden vision changes.

Discharging Physician: [Name], MD

Contact: [clinic phone] • **Nurse Navigator:** [name, phone]

Summary statement: Ms. Bennett remains in **complete remission** with **serial FLT3-ITD negativity**, has completed **HiDAC + midostaurin consolidation #4** without serious

AML080 — Diana Bennett (DOB 07.01.1963)

complications, and is discharged with clear monitoring and supportive plans while she continues her 1L program